João Coelho

Computer Science and Engineering Student · Lisbon, Portugal

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2nd-year Computer Science and Engineering student at Instituto Superior Técnico, driven by solving problems across the entire software stack. Building on a solid foundation in C, Java and Python, with a strong focus on systems programming and software architecture. Looking for a summer internship to build functional and scalable software.

EDUCATION

BSc in Computer Science and Engineering (LEIC-A)

2025 – 2028 (expected)

Instituto Superior Técnico - University of Lisbon — *Lisbon, Portugal*

- **Coursework:** Systems Programming, Object-Oriented Programming, Data Structures & Algorithms, Computer Architecture, Artificial Intelligence & Mathematics.

High School Diploma in Science and Technology

2022 – 2025

Escola Secundária Vergílio Ferreira

- **Academic Distinction:** Awarded Honor Roll in the final year for academic excellence.

PROJECTS

Express Retail Platform (ERP) (Ongoing)

Instituto Superior Técnico — Java · OOP · CLI · UML

- Building an object-oriented command-line retail management system featuring dynamic order splitting, inventory tracking and state serialization.
- Designing a highly extensible architecture following the Open-Closed Principle.

CloudIST VM Simulator (Ongoing)

Instituto Superior Técnico — C · Multithreading · Linux · POSIX

- Developing a data center simulator in C to manage virtual machine allocation and concurrent request processing.
- Implementing multithreading, process isolation (fork, exec), and strict resource limitations (CPU/Memory) via POSIX system calls.

Invoicing & Inventory CLI

Instituto Superior Técnico — C · Data Structures · Algorithms · CLI

- Built a stock-management system in C featuring a product catalogue, shopping cart, and invoice history.
- Implemented a custom merge sort, EAN checksum validation, and a wildcard search engine.

Scrabble Game & Agent

Instituto Superior Técnico — Python · Algorithms · AI

- Built a full Scrabble engine in Python for human and bot players across three difficulty levels.
- Implemented board/move validation, a custom xorshift shuffle algorithm, and a move-search AI agent.

My portfolio & CV

Personal Project — HTML · CSS · Web · GitHub Pages

- Developed a custom, responsive single-page portfolio and interactive CV.
- Designed a clean UI featuring seamless dark mode integration and a print-optimized layout.

SKILLS

Programming C, Java, Python, Assembly (RISC-V), HTML/CSS, Prolog

Tools & Environments Linux, Docker, Git, macOS / WSL

Languages Portuguese (Native), English (Fluent – EFSET C1), French (Elementary – DELF A2)

EXTRACURRICULARS & INITIATIVES

Early Coding Competitions & Hackathons

- **1st Place – Nova FCT Coding Fest (2016):** Won a collaborative hackathon at NOVA FCT, showing early problem-solving and teamwork.
- **Mobile Translation App – Happy Code Academy (2018):** Built a Portuguese-to-English Android translator with MIT App Inventor.